

CANOPY

prototype

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INTRODUCTION

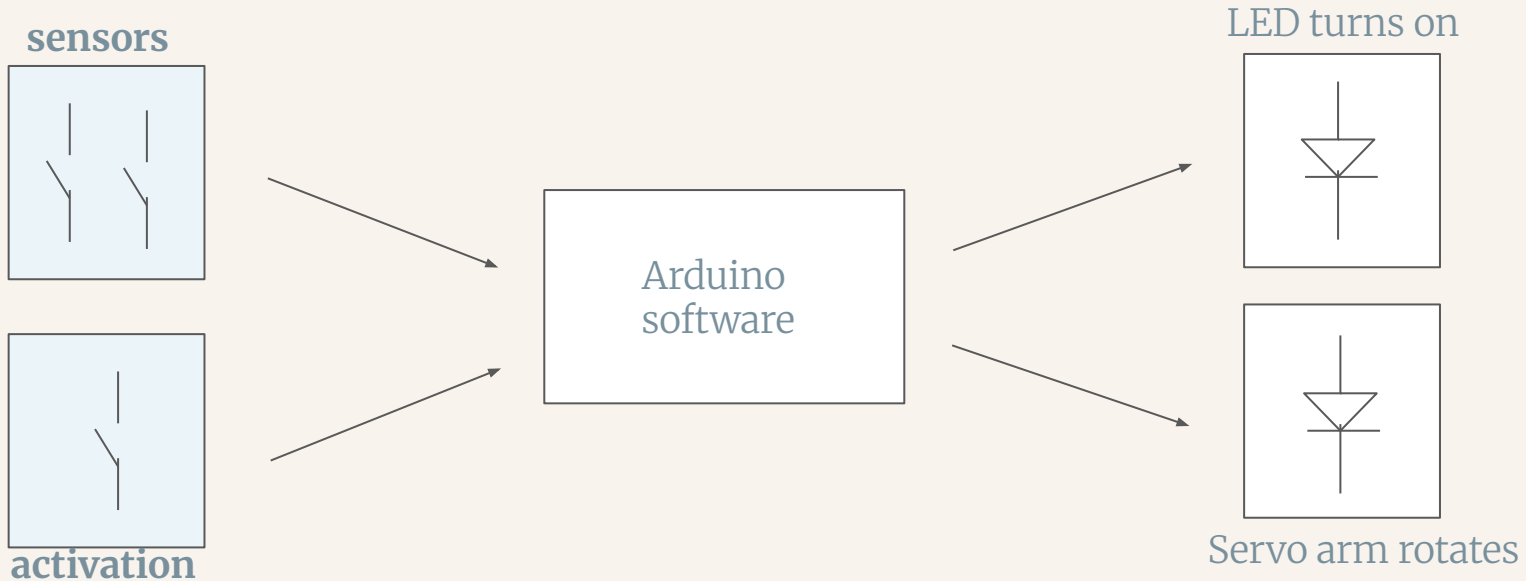
The purpose of our prototype is to have a canopy system deploy when there is “rain”.

For buildings like apartment complexes:

- LED activates to warn about impending weather
- Canopy deploys automatically when rain starts

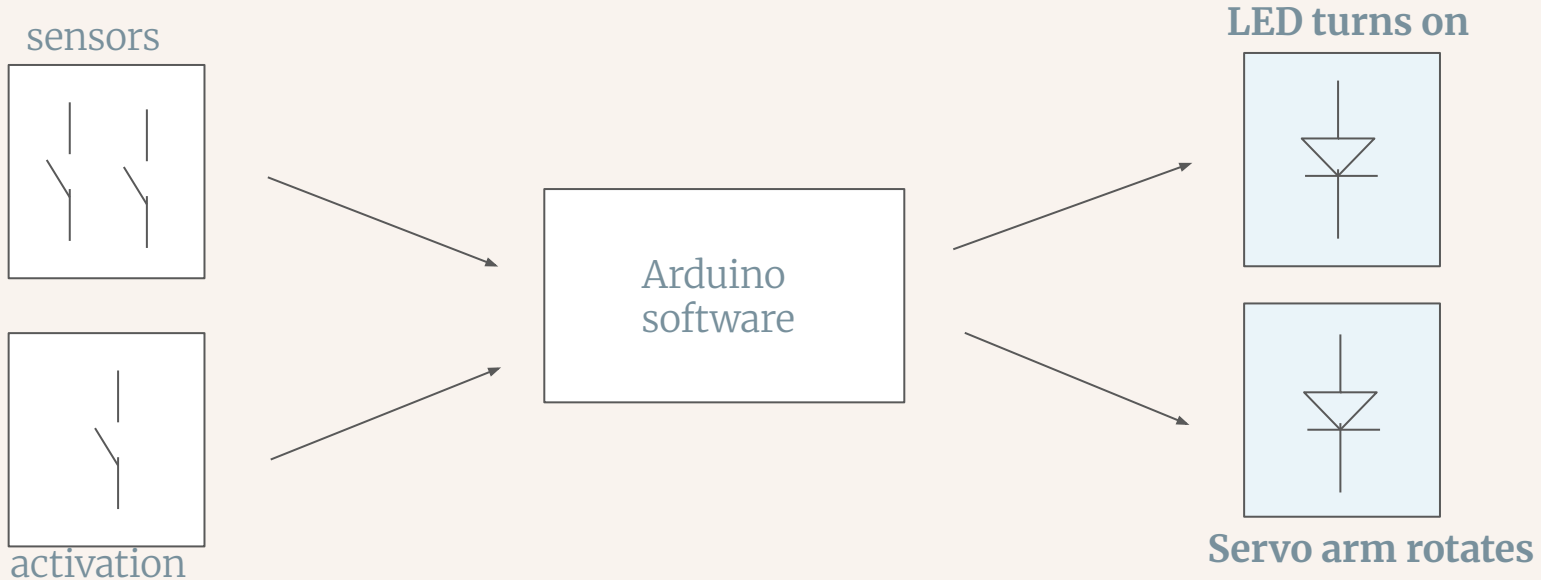
INPUTS

- Humidity is detected by homemade sensor
- “Rain” is detected by rain sensor
- Pressing the button



OUTPUTS

- System is activated
- LED is activated
- Servo arm rotates



CONSTRAINTS & REQUIREMENTS

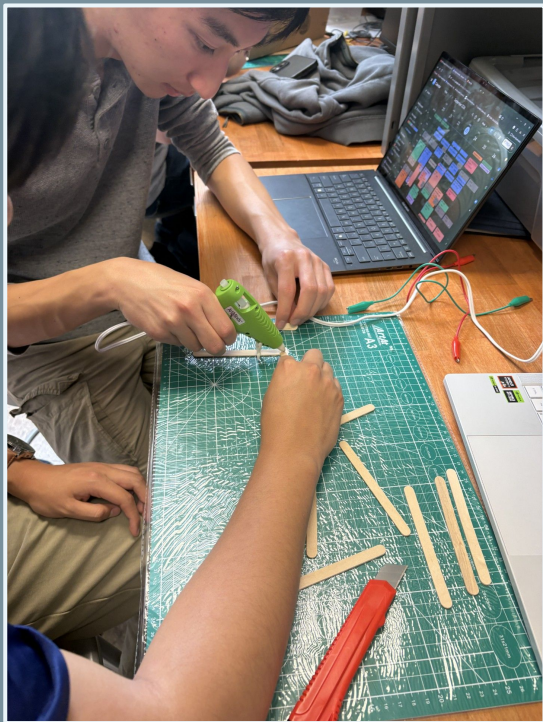
Constraints:

- System must function off 5 volts
- built out of provided materials
- meets specified peripherals
 - LED
 - Homemade sensor
 - Additional input element
 - motor

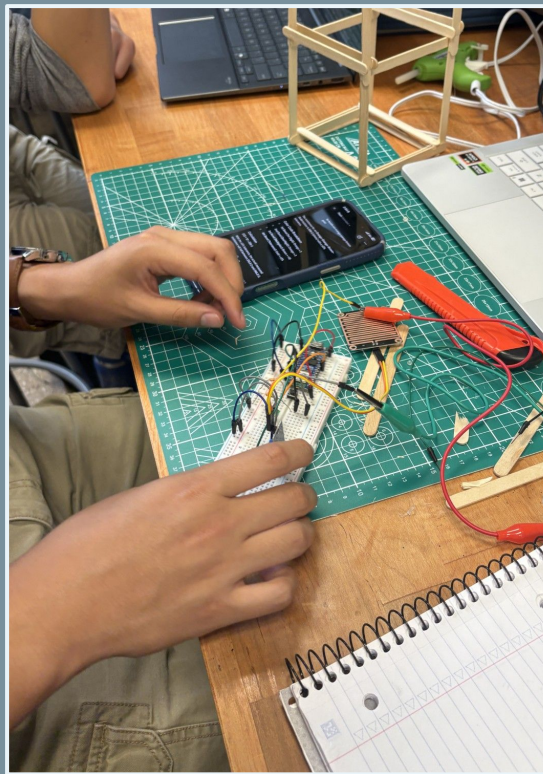
Requirements:

- Human interaction
- Provides a service/ support
- Multiple inputs & outputs
- Use of AI for code

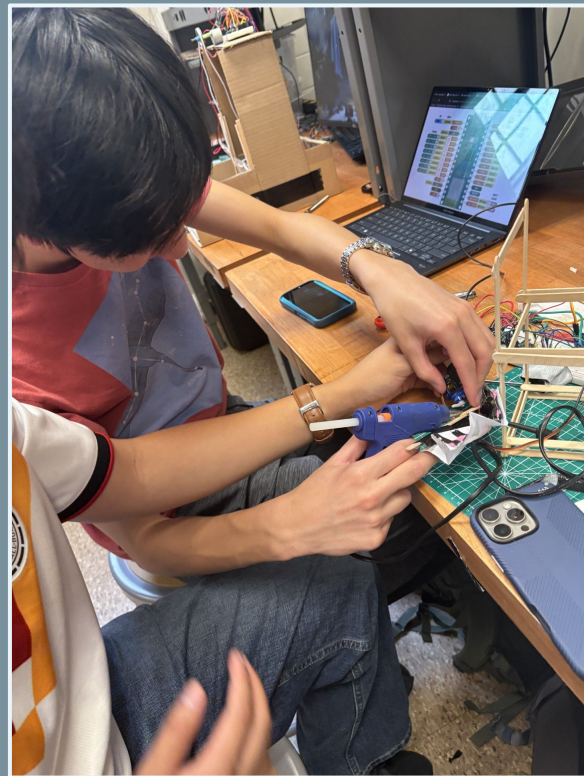
PROCESS



design



circuit wiring/ code



assembly

USE OF AI

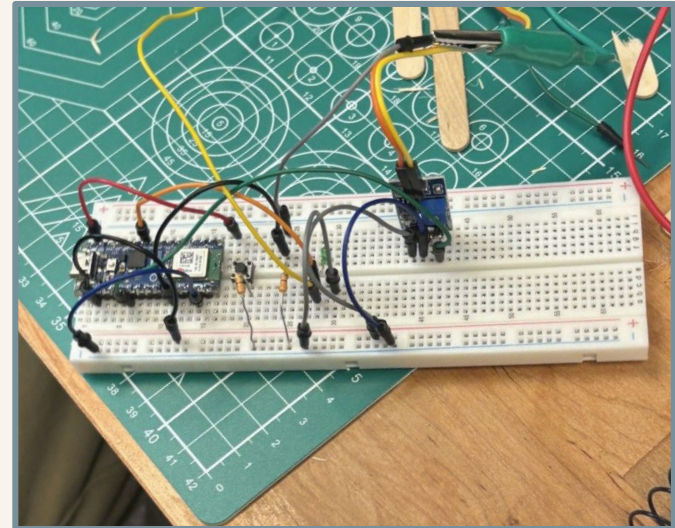
For the fabrication of our prototype we implemented AI in our process to develop the code for the arduino

```
# Hand-fabricated moisture sensor
# Connected to GPI014
moisture_sensor = ADC(Pin(14))

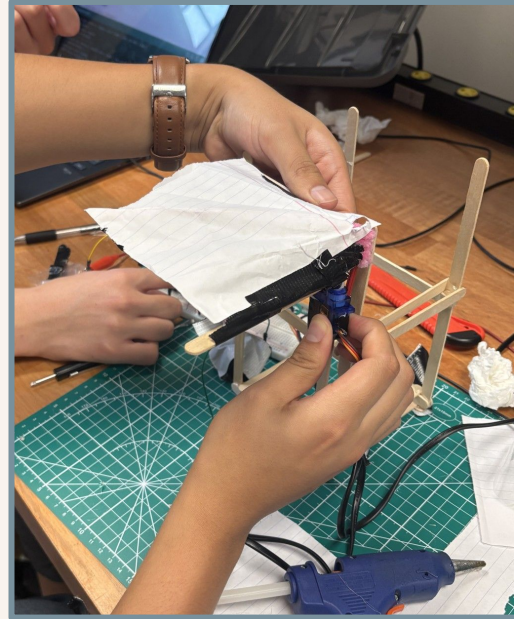
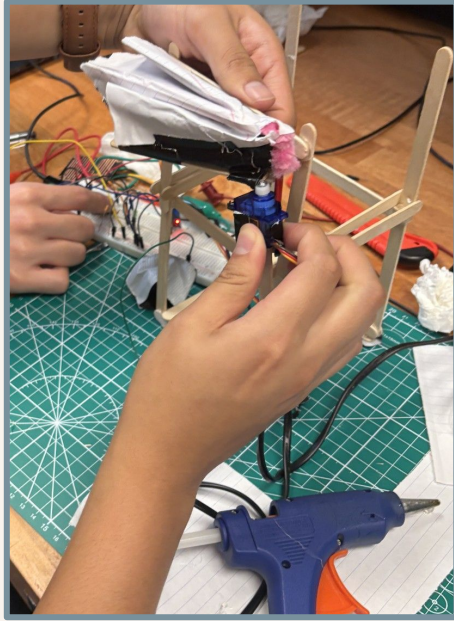
moisture_sensor atten(ADC.ATTN_11DB)
moisture_sensor.width(ADC.WIDTH_12BIT)

# Rain sensor
# Connected to GPI03
rain_sensor = ADC(Pin(3))

rain_sensor.atten(ADC.ATTN_11DB)
rain_sensor.width(ADC.WIDTH_12BIT)
```



RESULTS



Requirements:

- ☒ Human interaction
- ☒ Provides a service/ support
- ☒ Multiple inputs & outputs
- ☒ Use of AI for code

CONCLUSION

Our prototype fulfilled all **given requirements**

- Servo

Technical Issues:

- Awning storage position